

AI Emergency Patient Intake

Team Workflow and Role Ownership

Confirmed owners and team contribution update | v3 | 1 Sep 2026

Product goal

Reduce the information bottleneck between a distressed patient and emergency medical staff by converting natural conversation, caregiver input, staff observations and verified history into structured clinician-ready information.

ROLE 1 Voice AI Engineer Live audio, transcription, spoken replies, turn-taking and interruption handling. SHAHZAIB FRAZ ONE OWNER - CLEAR HANDOFFS	ROLE 2 AI / Agent Engineer prompts, conversation state, information extraction, agent decisions and safety guardrails. FAZWAN ZAINUDDIN ONE OWNER - CLEAR HANDOFFS	ROLE 3 Backend Engineer FastAPI, tool execution, authentication, workflow orchestration and external actions. v2: SSE publisher MOZZAM SHAHID ONE OWNER - CLEAR HANDOFFS	ROLE 4 Database & Integration Patient records, medical history, call logs, permissions, privacy and data retrieval. ISHAAN SAMA ONE OWNER - CLEAR HANDOFFS	ROLE 5 Frontend Engineer Live dashboard, structured summary, alerts, staff review and final approval interface. v2: SSE receiver JONATHAN ONE OWNER - CLEAR HANDOFFS
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1 SEP UPDATE - CONFIRMED OWNERS AND TEAM CONTRIBUTION

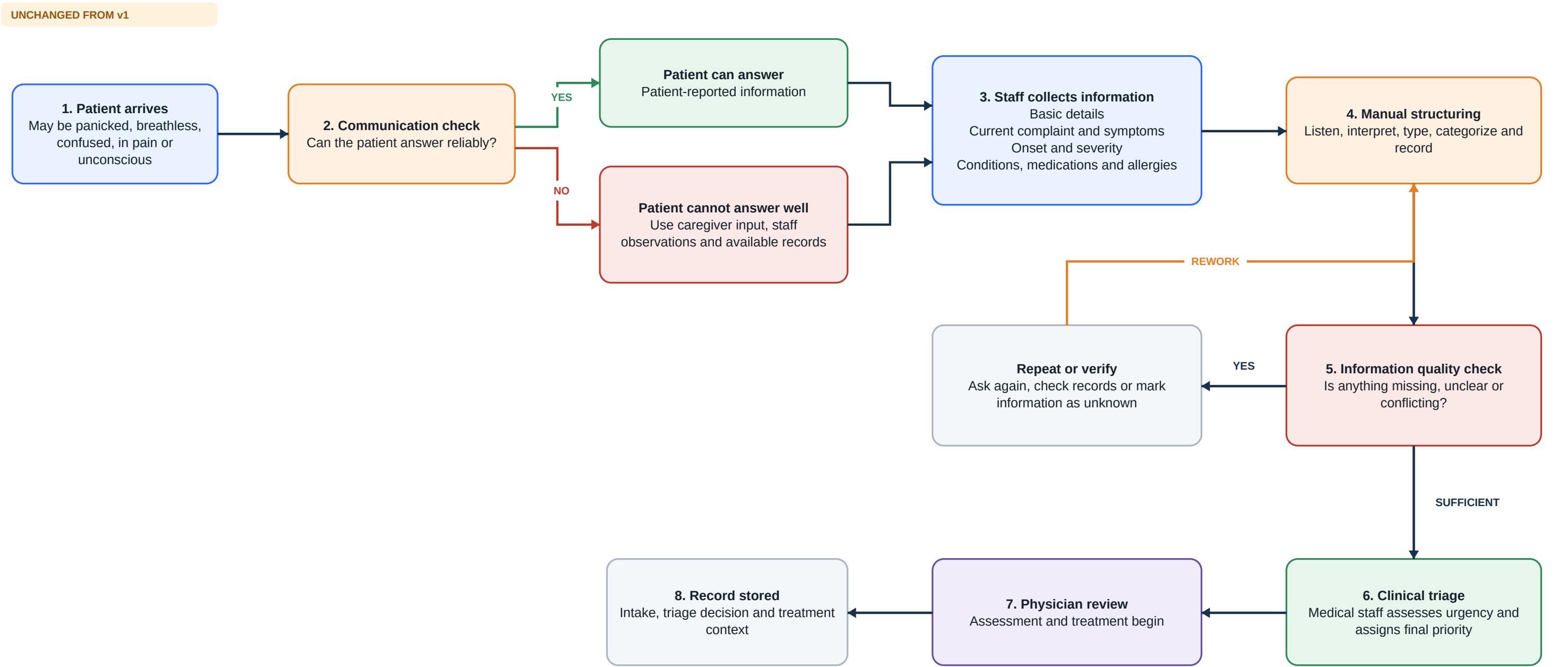
Confirmed: Shahzaib Fraz (Role 1), Fazwan Zainuddin (Role 2), Mozzam Shahid (Role 3), Ishaan Sama (Role 4) and Jonathan (Role 5).

Added: Ishaan Sama's EmergencyVoice tech-stack contribution is mapped to the team workflow on page 6. **Retained:** FastAPI + SSE and human-owned clinical decisions.

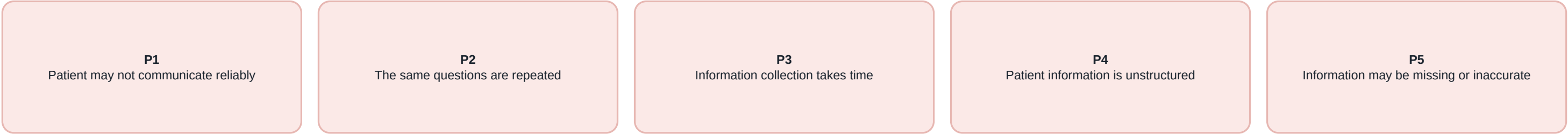
Team review: confirmed owners are named throughout. See page 6 for Ishaan Sama's technical contribution and open decisions.

1. Current Emergency Intake Flow

Where the communication and documentation bottlenecks occur



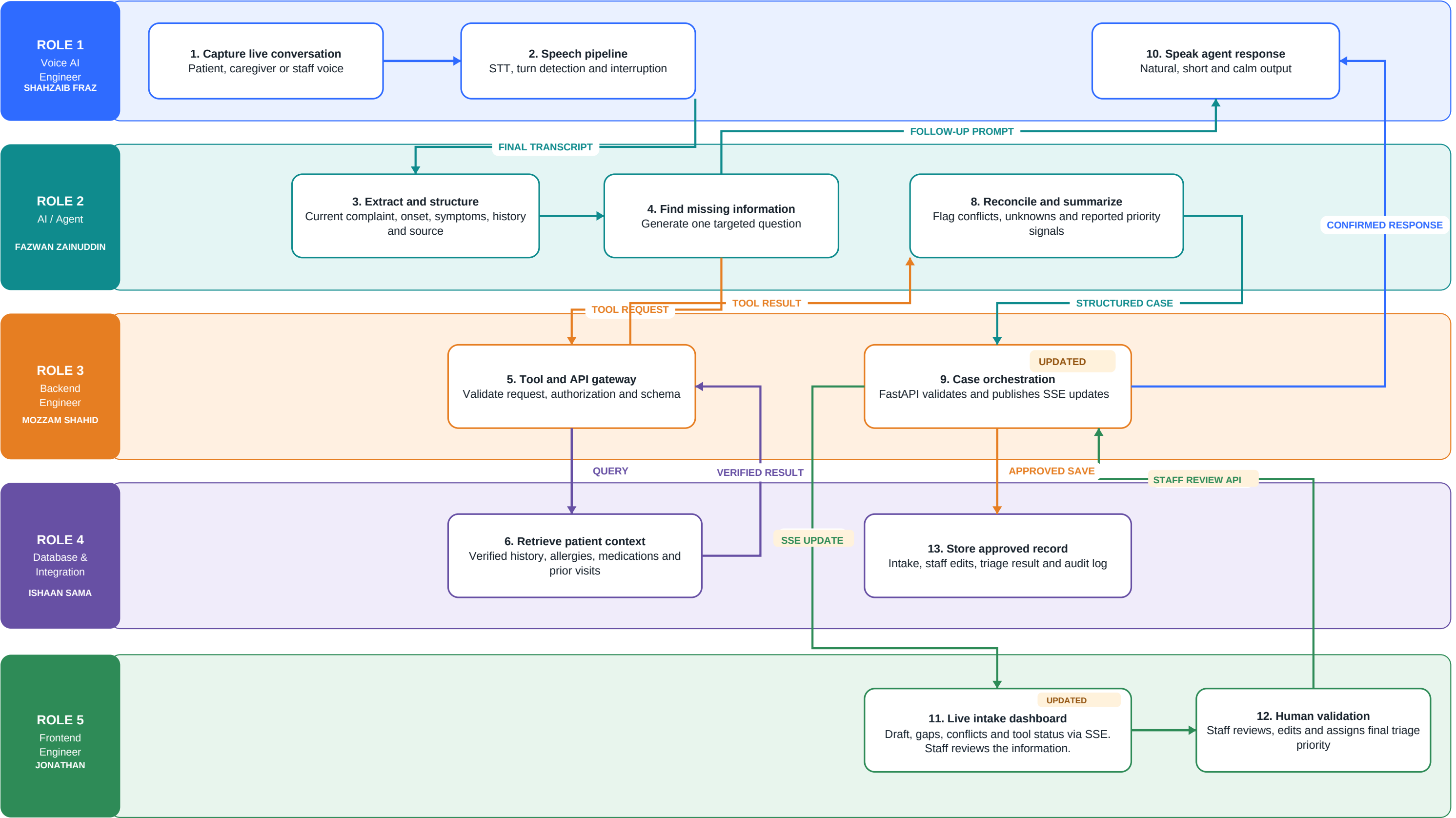
Pain points identified



2. Proposed AI-Assisted Workflow by Role

Each lane has one owner; arrows define the integration contracts

UPDATED IN v2: step 9 publishes SSE updates to step 11; staff review returns through an API. Lane order and step numbers are unchanged.



Human decision gate: AI structures and highlights reported information. Emergency medical staff remain responsible for clinical assessment and final triage priority.

3. Ownership and Handoff Contract

Use this page to prevent duplicate work and integration conflicts

Working rule: Every role owns its internal implementation, but all cross-role data must follow an agreed contract: event name, JSON schema, success response, error response and owner.

UPDATED CELLS Highlighted notes below clarify the SSE handoff. Role ownership and table structure are unchanged.

Role	Owns and delivers	Sends to other roles	Receives from other roles	Must not duplicate
Role 1 Voice AI Engineer SHAHZAIB FRAZ	AssemblyAI connection; microphone/audio streaming; STT/TTS; turn detection; interruptions; session state.	Final transcript turns, session ID/status and spoken-response completion.	Agent response text, conversation state and approved prompt configuration from Role 2.	Triage logic, backend business logic, database schema or dashboard UI.
Role 2 AI / Agent Engineer FAZWAN ZAINUDDIN	System prompt; conversation flow; extraction schema; missing-information logic; source/confidence labels; tool policy; safety guardrails; structured summary.	UPDATED v2 Response text to Role 1; tool requests and structured extraction to Role 3. The backend owns the SSE event wrapper.	Final transcript from Role 1; verified tool result from Role 3; retrieved context through backend tools.	Direct database access, API implementation, UI code or the medical staff's final triage decision.
Role 3 Backend Engineer MOZZAM SHAHID	UPDATED v2 FastAPI; authentication; validation; tool execution; workflow orchestration; SSE publishing and reconnection; staff-review APIs.	UPDATED v2 Queries and approved saves to Role 4; live SSE updates to Role 5; actual tool results to Role 2.	Validated tool requests from Role 2; data responses from Role 4; staff actions from Role 5.	Agent prompt/decision policy, physical data model ownership or frontend rendering.
Role 4 Database & Integration ISHAAN SAMA	Patient schema; synthetic records; history and allergies; call logs; audit trail; permissions; privacy; retention; query performance.	Verified records, not-found responses, conflicts and save confirmations to Role 3.	Authorized read/write requests from Role 3 and approved staff updates routed by backend.	Calling the LLM directly, frontend UI, voice pipeline or backend workflow decisions.
Role 5 Frontend Engineer JONATHAN	Live monitoring dashboard; structured intake; source/confidence display; missing/conflict flags; staff edit, review and approval interface.	Staff corrections, approval, final triage selection and action requests to Role 3.	UPDATED v2 SSE updates with the draft, missing information, conflicts and tool status from Role 3; confirmed review/save responses.	Direct database writes, independent triage logic, voice pipeline or backend tool execution.

Source: Emergency Patient Intake - Problem Analysis & AI Opportunity Assessment. This is an architecture proposal for team agreement, not a clinical protocol.

4. Team Status and Next Actions

NEW IN v2

1 September update - confirmed owners, explicit next actions and one shared synthetic-case target.

Current direction: FastAPI + SSE. **Example received:** EXTRACTION_UPDATE JSON. **Implementation status:** each owner to confirm; no completion is assumed.

The shared payload is an example, not yet the final contract. SSE specifies live delivery; it does not decide which database we use.

What changed from the version already shared?

Reference in v1	Earlier wording / position	v2 update
Page 3: step 9 to step 11	LIVE EVENT; transport not named on the diagram.	SSE UPDATE: FastAPI sends live draft updates to the staff dashboard.
Page 3: review return path	STAFF APPROVAL.	STAFF REVIEW API: staff edits/approval return to Role 3; approved records are saved via Role 4.
Page 4: handoffs; page 5 added	Generic live case events; no dated next-action sheet.	SSE sender/receiver responsibilities clarified; shared data decisions and next deliverables listed below.

Next deliverable from each owner

Same role / owner	Next action	Evidence to share with the team
Role 1 - Shahzaib Fraz	Provide transcript turns and session references; connect the follow-up response from Role 2.	One synthetic spoken exchange becomes text, followed by one question and answer.
Role 2 - Fazwan Zainuddin	Align extraction fields with the sample; retain source labels; prepare the prompt and expected outputs.	Examples for sufficient, missing and conflicting information; output ready for Role 3 validation.
Role 3 - Mozzam Shahid	Confirm intake/review interfaces and authentication; publish the agreed sample through SSE.	One sample update reaches Role 5; reconnect works; review and save confirmation are returned.
Role 4 - Ishaan Sama	Confirm database, identifiers, authorised retrieval and approved-record/audit storage.	One synthetic lookup, one not-found result and one saved approved record without duplicate writes.
Role 5 - Jonathan	Display SSE draft updates; show gaps/conflicts; send staff edits and approval back to Role 3.	One live update without refresh; staff corrections survive later updates; save status is visible.
Demo - Mariam Habib	Coordinate a script and joint test once owners report readiness; use synthetic data only.	One complete intake-to-review-to-save demonstration; any simulated component is labelled.

TEAM CONFIRMATION BEFORE INTEGRATION

Roles 2 / 3 / 5: agree field/event names, full-state vs partial updates, versions, unknown/source handling and review protection. **Roles 3 / 4 / 5:** confirm database, authentication and reconnect/save behaviour.

Everyone: reply with agree / requested changes, what is ready, next deliverable, blocker and target test date. First target: one shared synthetic case reaches the dashboard, is reviewed by staff and is saved successfully.

Scope: same workflow and owners as the previously shared PDF. AI prepares a draft; authorised staff retain the final clinical decision. Clinical approval rules require domain review.

Technical reference: [FastAPI SSE documentation](#). Based on the 30 Aug workflow and the 31 Aug team update. Full implementation readiness is for owners to confirm.

5. Confirmed Team Roles and Ishaan Sama Contribution

Registered team names are fixed below. Readiness and delivery status remain for each owner to report.

NEW IN v3

Confirmed role ownership

Role	Registered team member	Primary ownership
Role 1	Shahzaib Fraz	Real-time voice, transcript turns, turn-taking, interruption handling and spoken response.
Role 2	Fazwan Zainuddin	Agent prompt, structured extraction, source/confidence, gaps, conflicts and safety guardrails.
Role 3	Mozzam Shahid	FastAPI backend, authentication, validation, orchestration, SSE and review APIs.
Role 4	Ishaan Sama	Synthetic patient data, authorised retrieval, persistence, audit, privacy and retention.
Role 5	Jonathan	Staff dashboard, live updates, human review, corrections and approval interface.
Demo	Mariam Habib	Team coordination, synthetic demo script, rehearsal and presentation flow.

Ishaan Sama tech-stack input - incorporated into team planning

Contribution	How it is incorporated	Owner / status
AssemblyAI voice layer	Voice Agent API with STT, VAD/turn-taking, sentiment, disfluency, word timestamps and tool calling is the proposed real-time signal source.	Role 1 + Role 3; model/API choice to confirm.
Demo call transport	Use browser/WebRTC for the fastest demo; keep Twilio as an optional realism upgrade if time and cost permit.	Role 1 + Role 3; open decision.
Schema-first extraction	Use JSON-Schema tool calling; represent missing data as nullable/unknown and never force or guess a clinical value.	Role 2 + Role 3; aligned with current contract.
Distress indicator	Combine sentiment, disfluency and pacing into an auditable HIGH/MEDIUM/LOW signal, kept separate from clinical severity and final triage.	Roles 2/3/5; formula requires agreement.
Mock patient data	Use synthetic records, server-side filtered lookup and an explicit retention/audit policy. SQLite is sufficient for MVP; Postgres remains an option.	Role 4 + Role 3; DB choice open.
Brief and dashboard	Generate a fixed structured brief, publish through the current FastAPI + SSE path and require visible human review. LeMUR is a candidate, not an automatic dependency.	Roles 2/3/5; FastAPI + SSE retained.
Privacy and scope	Apply PII redaction to retained transcripts, use synthetic data, state what is deleted/retained and prohibit autonomous diagnosis or triage scoring.	All roles; mandatory safety constraint.

DECISIONS TO CLOSE EARLY

Universal-3 Pro/API availability; browser-only vs Twilio; SQLite vs Postgres; LeMUR vs current LLM path; exact distress-signal formula; raw-audio and transcript retention.

Contribution source: Ishaan Sama, EmergencyVoice - Tech Stack Document (31 Aug 2026). Team direction remains FastAPI + SSE unless the owners agree a documented change.